STAT 412: Case Study 2

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Introduction

We wish to be able to find a way to assess the helpfulness of certain filler words for identifying the author of the disputed Federalist papers. Since it is generally agreed upon that the disputed papers were not written by Jay, we will assess non-disputed Federalist Papers written by Madison or Hamilton to determine the usefulness of certain filler words. Depending upon what a paper is about, it could use a specific word a lot depending on the context of the article, which is why we only consider "filler words" in the analysis. We will assess how helpful six filler words (an, of, the, is, and, to) are for solving the research question as they occur in every non-disputed federalist paper.

0.1 Research Question

Which (if any) filler words might be useful for helping solve the authorship problem? Which (if any) filler words are not helpful?

1 Model description

We will use a negative binomial distribution to model the frequency of a specific word in a specific article. Y_{jik} represents the sampling model for the number of appearances of word j in article i written by author k:

$$Y_{jik}|\alpha_{jk}, p_{jik} \overset{ind}{\sim} NegBinom(\alpha_{jk}, p_{jik})$$

$$i = \{1, 2, ..., 68\}$$

for the 68 undisputed articles written by Madison or Hamilton.

$$j = \{1, 2, 3, 4, 5, 6\}$$

for the 6 selected words to investigate

$$k = \{1, 2\}$$

for the 2 authors we are analyzing (Madison and Hamilton).

α_{jk} and β_{jk} alone are not easily interpretable because there is no notion of trials until a certain number of successes when considering the frequency of a word in an article. This means we cannot use the typical interpretation of the parameters of a negative binomial distribution which would model the number of trials until the α success where the probability of success is p_{jik} in our case. Instead, we will consider the mean of the distribution.

$$E[Y_{jik}|\alpha_{jk}, p_{jik}] = \frac{\alpha_{jk}(1 - p_{jik})}{p_{jik}}$$

We will reparameterize p_{jik} as $p_{jik} = \frac{\beta}{\beta + \frac{n_i}{1000}}$ to give us

$$E[Y_{jik}|\alpha_{jk}, p_{jik}] = \frac{\alpha_{jk}(1 - \frac{\beta_{jk}}{\beta_{jk} + \frac{n_i}{1000}})}{\frac{\beta_{jk}}{\beta_{jk} + \frac{n_i}{1000}}}$$

With some simplification we arrive at

$$E[Y_{jik}|\alpha_{jk}, p_{jik}] = (\frac{\alpha_{jk}}{\beta_{jk}})(\frac{n_i}{1000})$$

Now we can see why it is useful to reparameterize p_{jik} in terms of β_{jk}. Since the expected value of Y_{jik} represents the expected number of times we expect to see word j in article i, and $E[Y_{jik}|\alpha_{jk}, p_{jik}] = (\frac{\alpha_{jk}}{\beta_{jk}})(\frac{n_i}{1000})$, we can interpret $\frac{\alpha_{jk}}{\beta_{jk}}$ as the number of times we expect the word to show up per 1000 words in one of the papers. We care about this value because we want to assess a word's helpfulness in distinguishing between a paper written by the two authors, so if we find a significant difference in the rate of use of certain words between Madison and Hamilton, then these words are likely helpful in identifying the author of disputed articles.

1.1 Assumptions

We are given the task to model this data with a Negative Binomial distribution but this is likely the case because this distribution assumes that there is discrete and high variance count data. This claim of high variance count data is what makes the negative binomial the more appropriate model to choose rather than the poisson because the negative binomial allows for the variance to be greater than the mean (a very limiting assumption made by the poisson distribution). The counts Y_{jik} (for a certain word and author) are independent of each other, conditional on the parameters α_{jk} and β_{jk}. This assumes that the use of a word in one article does not directly influence its use in another. Furthermore, an important assumption is that the author's use of filler words is a consistent writing style, constant across all the different federalist papers.

1.2 Methods for Answering Research Question

As stated in the introduction, we want to assess a word's helpfulness in solving the authorship problem. In order to do so, we will analyze the posterior distributions of the rate of use of the six specified filler words between Madison and Hamilton. As discussed above, the rate of use of a word for an author per 1000 words is $\frac{\alpha_{jk}}{\beta_{jk}}$. Therefore, we will approximate posterior distributions of $\frac{\alpha_{jk}}{\beta_{jk}}$ for Madison and Hamilton for the 6 different words. Then, we can analyze the difference in distributions of $\frac{\alpha_{jk}}{\beta_{jk}}$ for a given word between Madison and Hamilton. If we find that, for a given word, the distribution of use of that word is significantly different between Madison and Hamilton in that there is no overlap in the 95% credible intervals, that is an indicator that the word is likely helpful in solving the authorship problem. On the other hand, if we find that the posterior densities of the expected usage rate of a given word are very similar between Madison and Hamilton in that their 95% credible intervals overlap, then the word is likely not very helpful in solving the authorship problem because the rate of use of that word will not be much more likely to fall in one distribution than the other.

2 Implementation

As discussed above, the research question will be answered by analyzing $f(\frac{\alpha_{jk}}{\beta_{jk}}|\vec{y})$. This posterior distribution will be obtained with a Metropolis sampling method. In order to use a Metropolis sampling method, we need priors for the parameters that we want posteriors for. Since α_{jk} and $\frac{\beta_{jk}}{\beta_{jk}+\frac{n_i}{1000}}$ are parameters for a negative binomial distribution, α_{jk} and β_{jk} must be positive real numbers. Given the parameter space of α_{jk} and β_{jk}, we will use gamma distributions as priors for both of them (the thought behind the parameters for the priors is after the description of the Metropolis algorithm). We also need to consider the fact that in order to use a Metropolis sampling method, we need to sample a proposal for the random variable on every iteration from a symmetric distribution. We chose to use a normal distribution to sample the proposals for α_{jk} and β_{jk}. We ran into issues because the support of a normal distribution is $(-\infty, \infty)$ leading us to end up with negative values for α_{jk} and β_{jk} which then cannot be used as parameters for a negative binomial distribution. To address this issue, we performed a change of variables for both α_{jk} and β_{jk}. In order to ensure that α_{jk} and β_{jk} both stay positive, we perform a log transformation on them. In doing so, we set $Z = log(\alpha_{jk})$ and $\omega = log(\beta_{jk})$. Then, we notice the following:

$\alpha_{jk} = e^{Z_{jk}}$ and $\beta = e^{\omega_{jk}}$

And

$f(z) = f_{\alpha_{jk}}(e^{Z_{jk}})|e^{Z_{jk}}| = f_{\alpha_{jk}}(\alpha_{jk})\alpha_{jk}$ and $f(\omega) = f_{\beta_{jk}}(e^{\omega_{jk}})|e^{\omega_{jk}}| = f_{\beta_{jk}}(\beta_{jk})\beta_{jk}$

Therefore we can sample a proposed Z_{jk} from a normal centered at $log(\alpha_{jk})$ and sample a proposed ω_{jk} from a normal centered at $log(\beta_{jk})$. So $Z_{jk} \sim N(log(\alpha_{jk}), \delta_{jk})$ and $\omega_{jk} \sim N(log(\beta_{jk}), \delta_{jk})$, where α_{jk} and β_{jk} are the most recently accepted parameters and δ_{jk} determines how quickly we allow the algorithm to explore the parameter space. Then we can find a proposed α_{jk} and β_{jk} because $\alpha_{jk} = e^{Z_{jk}}$ and $\beta_{jk} = e^{\omega_{jk}}$. This way, α_{jk} and β_{jk} are guaranteed to be positive. Then, all we have to do is substitute $f_{\alpha_{jk}}(\alpha_{jk})\alpha_{jk}$ and $f_{\beta_{jk}}(\beta_{jk})\beta_{jk}$ in for $f_{z_{jk}}(z_{jk})$ and $f_{\omega_{jk}}(\omega_{jk})$, respectively in the log acceptance ratio (r_α and r_β) leaving us with acceptance ratios (on the log scale) of:

For the remainder of the report, when referring to α and β the subscripts are implied.

$$\begin{aligned}
r_\alpha = &\sum_{i=0}^{68} \log\left(\binom{y_{jik} + \alpha_{\text{prop}} - 1}{y_{jik}} p_{jik}^{\alpha_{\text{prop}}} (1 - p_{jik})^{y_{jik}}\right) \\
&- \sum_{i=0}^{68} \log\left(\binom{y_{jik} + \alpha - 1}{y_{jik}} p_{jik}^{\alpha} (1 - p_{jik})^{y_{jik}}\right) \\
&- \log(\alpha^{a_\alpha - 1} e^{-b_\alpha \alpha}) + \log(\alpha_{\text{prop}}^{a_\alpha - 1} e^{-b_\alpha \alpha_{\text{prop}}}) \\
&+ (\log(\alpha_{\text{prop}}) - \log(\alpha))
\end{aligned}$$

and

$$\begin{aligned}
r_\beta = &\sum_{i=0}^{68} \log\left(\binom{y_{jik} + \alpha - 1}{y_{jik}} p_{jik_{\text{prop}}}^{\alpha} (1 - p_{jik_{\text{prop}}})^{y_{jik}}\right) \\
&- \sum_{i=0}^{68} \log\left(\binom{y_{jik} + \alpha - 1}{y_{jik}} p_{jik}^{\alpha} (1 - p_{jik})^{y_{jik}}\right) \\
&+ \log(\beta_{\text{prop}}^{a_\beta - 1} e^{-b_\beta \beta_{\text{prop}}}) - \log(\beta^{a_\beta - 1} e^{-b_\beta \beta}) \\
&+ (\log(\beta_{\text{prop}}) - \log(\beta))
\end{aligned}$$

All in all, for every combination of author and word there is a corresponding posterior distribution for $\frac{\alpha}{\beta}$. The means that we will obtain twelve posterior distributions in total: two (one for each author) for each of the six words.

We needed to select priors for α and β that reflect our prior beliefs about how frequently the author will use the word. Since $\frac{\alpha}{\beta}$ is interpretable and we can state our beliefs about how frequently certain words will show up in a paper per 1000 words, in order to pick priors for α and β we will look at the induced distribution of $\frac{\alpha}{\beta}$ with priors of $\alpha \sim Gamma(a_\alpha, b_\alpha)$ and $\beta \sim Gamma(a_\beta, b_\beta)$ for each of the six words.

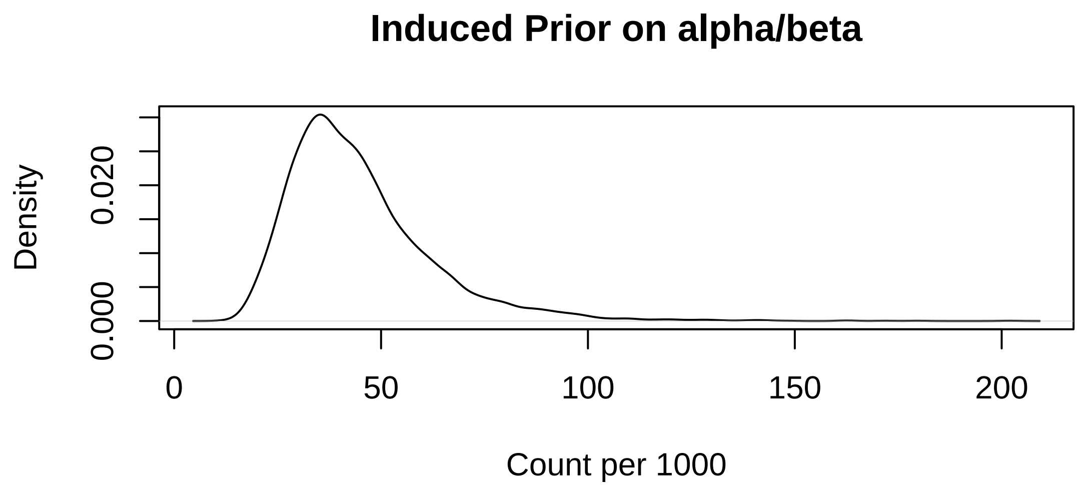

Figure 1: Induced prior on alpha/beta which represents our prior beliefs on an expected use of a filler word per 1000 words

Our induced prior distributions of $\frac{\alpha}{\beta}$ (Figure 1) for each of 6 different words are weakly informative Gamma priors that are identical for all filler words. The $\alpha \sim Gamma(40, 1)$ and $\beta \sim Gamma(10, 10)$ priors were chosen to have a wide variance such that it encompasses what we expect to be the frequency range of the filler words. This single weakly informative prior approach reflects our initial uncertainty about word frequencies while allowing for the posterior distributions to be "dominated" by the data allowing the data to speak.

2.1 MCMC code

```
read_dat <- function(wrd, prsn){

  filler_word_counts <- all_dat|>
    filter(Authorship == prsn)|>
    filter(Disputed == "no")|>
    filter(word == wrd)|>
    pull(N)

  total_word_counts <- all_dat|>
    filter(Authorship == prsn)|>
    filter(Disputed == "no")|>
    filter(word == wrd)|>
    pull(Total)

  return_list <- list(filler_count = filler_word_counts,
                      total_count = total_word_counts)
```

```
  return(return_list)
}

sample_post <- function(wrd, author, a_a, b_a, a_b, b_b, delt_z, delt_y) {

  set.seed(33)

  S <- 20000

  a_a <- a_a
  b_a <- b_a

  a_b <- a_b
  b_b <- b_b

  ys <- read_dat(wrd, author)$filler_count
  ns <- read_dat(wrd, author)$total_count

  beta <- 100
  alpha <- 1
  delta_z <- delt_z
  delta_y <- delt_y
  p_prop <- 1/10
  p <- beta/(beta+(ns/1000))

  POST <- matrix(NA, nrow = S, ncol= 3)
  acc_matrix <- matrix(0, nrow = S, ncol = 2)

  for(s in 1:S){

    # propose
    z_prop <- rnorm(1, log(alpha), delta_z)
    alpha_prop <- exp(z_prop)
    alpha_r <- sum(dnbinom(ys, alpha_prop, p, log = T)) -
      (sum(dnbinom(ys, alpha, p, log = T))) +
      dgamma(alpha_prop, a_a, b_a, log = T) -
      dgamma(alpha, a_a, b_a, log = T)+
      (log(alpha_prop) - log(alpha))
    # accept or reject
    if(log(runif(1)) < alpha_r){
      alpha <- alpha_prop
      acc_matrix[s,1] <- 1
    }
    # propose
```

```
    y_prop <- rnorm(1, log(beta), delta_y)
    beta_prop <- exp(y_prop)
    p_prop <- beta_prop/(beta_prop+(ns/1000))

    beta_r <- sum(dnbinom(ys, alpha, p_prop, log = T)) -
    (sum(dnbinom(ys, alpha, p, log = T))) +
    dgamma(beta_prop, a_b, b_b, log = T) -
    dgamma(beta, a_b, b_b, log = T) +
    (log(beta_prop) - log(beta))
    # accept or reject
    if(log(runif(1)) < beta_r){
      beta <- beta_prop
      p <- p_prop
      acc_matrix[s,2] <- 1
    }
    # store
    POST[s,] <- c(alpha, beta, alpha/beta)
  }

  accept_a <- mean(acc_matrix[,1])
  accept_b <- mean(acc_matrix[,2])
  POST_BURNED <- POST[-c(1:(S/2)), ]
  return_list <- list(a_sam = POST_BURNED[,1], b_sam = POST_BURNED[,2],
                      ab_sam = POST_BURNED[,3], accept_a = mean(acc_matrix[,1]),
                      accept_b = mean(acc_matrix[,2]))
  return(return_list)
}
```

2.2 Diagnostics

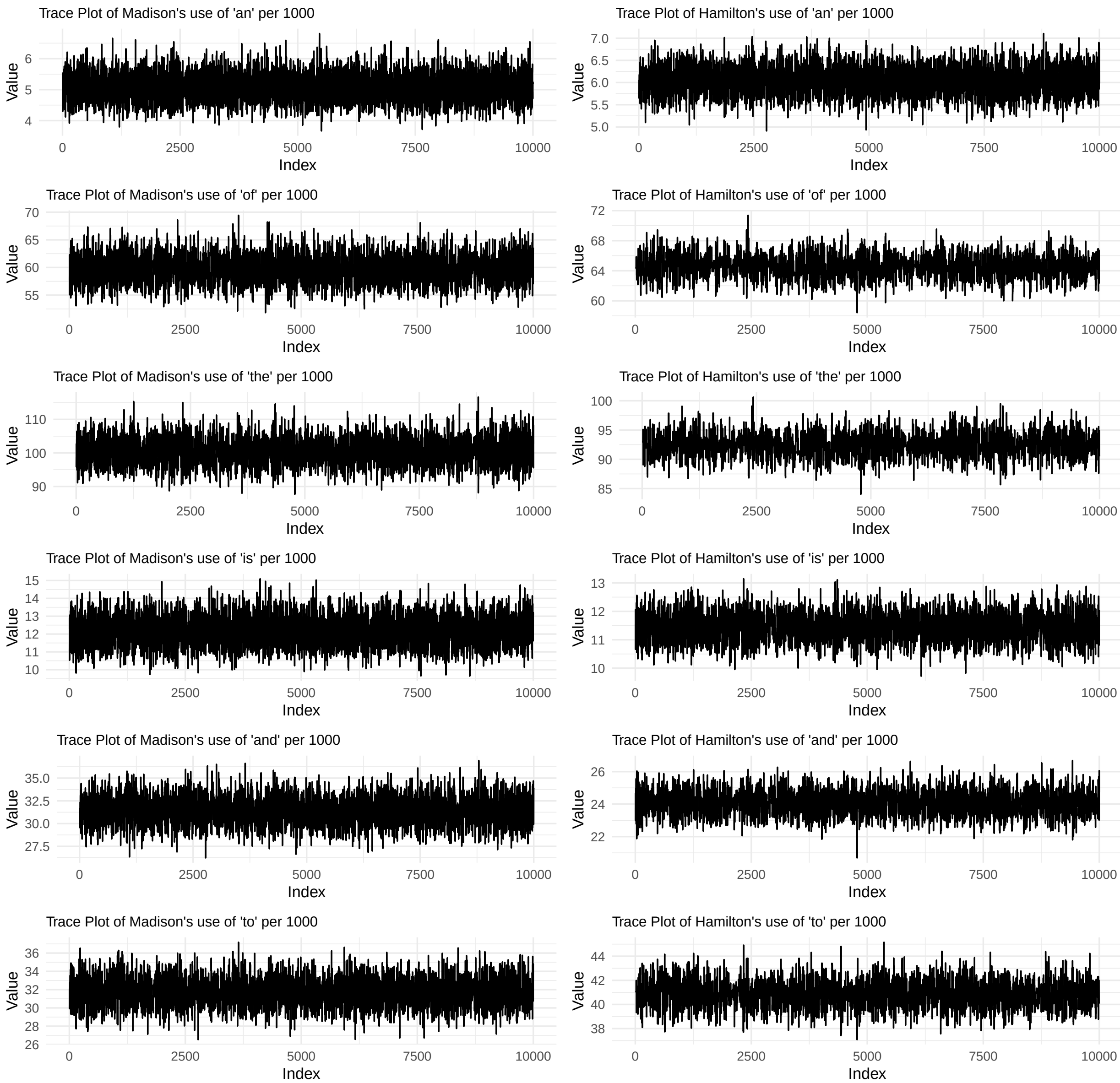

Figure 2. Trace plots of the metropolis algorithm of alpha/beta for each of the chosen words and each author Madison and Hamilton, on the left and right respectivly, confirming MCMC convergence.

The trace plots (Figure 2) illustrate that the metropolis MCMC has converged, as the chains explore the parameter space without "getting stuck" and thus are stationary.

Table 1: Acceptance Rates for Each Author for Each of the 6 Words

Word	Acceptance Rate alpha Madison	Acceptance Rate alpha Hamilton	Acceptance Rate beta Madison	Acceptance Rate beta Hamilton
an	0.435	0.291	0.440	0.294
of	0.243	0.143	0.248	0.147
the	0.234	0.134	0.239	0.139
is	0.356	0.248	0.365	0.246
and	0.274	0.187	0.284	0.191
to	0.273	0.162	0.280	0.168

As seen by Table 1 our metropolis algorithm's acceptance ratios are all reasonable (between 0.1 and 0.45) being in this "optimal" range allows for our algorithm to explore well but also takes reasonable sized steps in that it is not always always accepting (proposed steps are too small) or always rejecting (proposed steps are too large).

Table 2: Effective Sample Size by Word and Author

Word	ESS(alpha/beta) Madison	ESS(alpha/beta) Hamilton
an	3938.4	3753.5
of	3312.3	1915.5
the	2877.9	1814.6
is	4488.3	3011.9
and	3507.8	2112.7
to	3531.6	2074.2

The effective sample sizes (Table 2) are not close to the number of post burn-in samples. This is likely because the lower effective size correlates with the word and author with low acceptance rate. A lower acceptance rate means that the model rejects the proposed parameter more often. When the proposal is rejected, the parameter remains the same as the previous iteration, which obvoiusly leads to a high positive autocorrelation and low effective sample size. Increasing the number of iterations to run the metropolis MCMC allows us to get an effective sample size which is of adequate size to perform Bayesian analysis on.

Let's look at some posterior predictive checks to assess model fit.

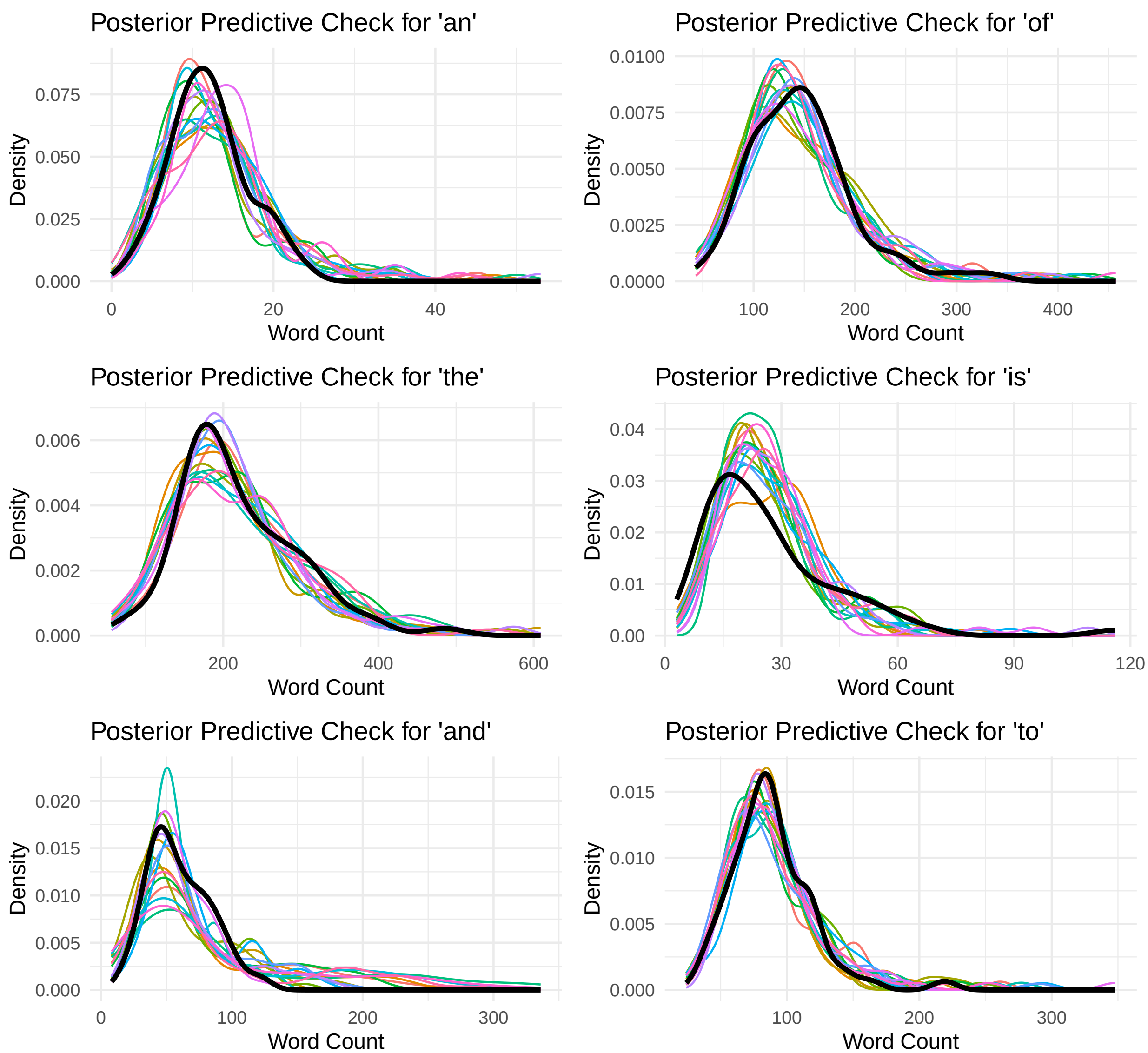

Figure 3. In these plots, the colored lines represent posterior predictive samples from our model for the use of the given word and the black line represents the density of actual use of that that word as observed in the data for both Madison and Hamilton.

The model does a very good job at approximating the distribution of the use of each the words because the center and shape of the posterior predictive densities closely resemble the center and shape of the observed data. Therefore we will proceed with the analysis under the use of this model (Figure 3).

3 Results and Findings

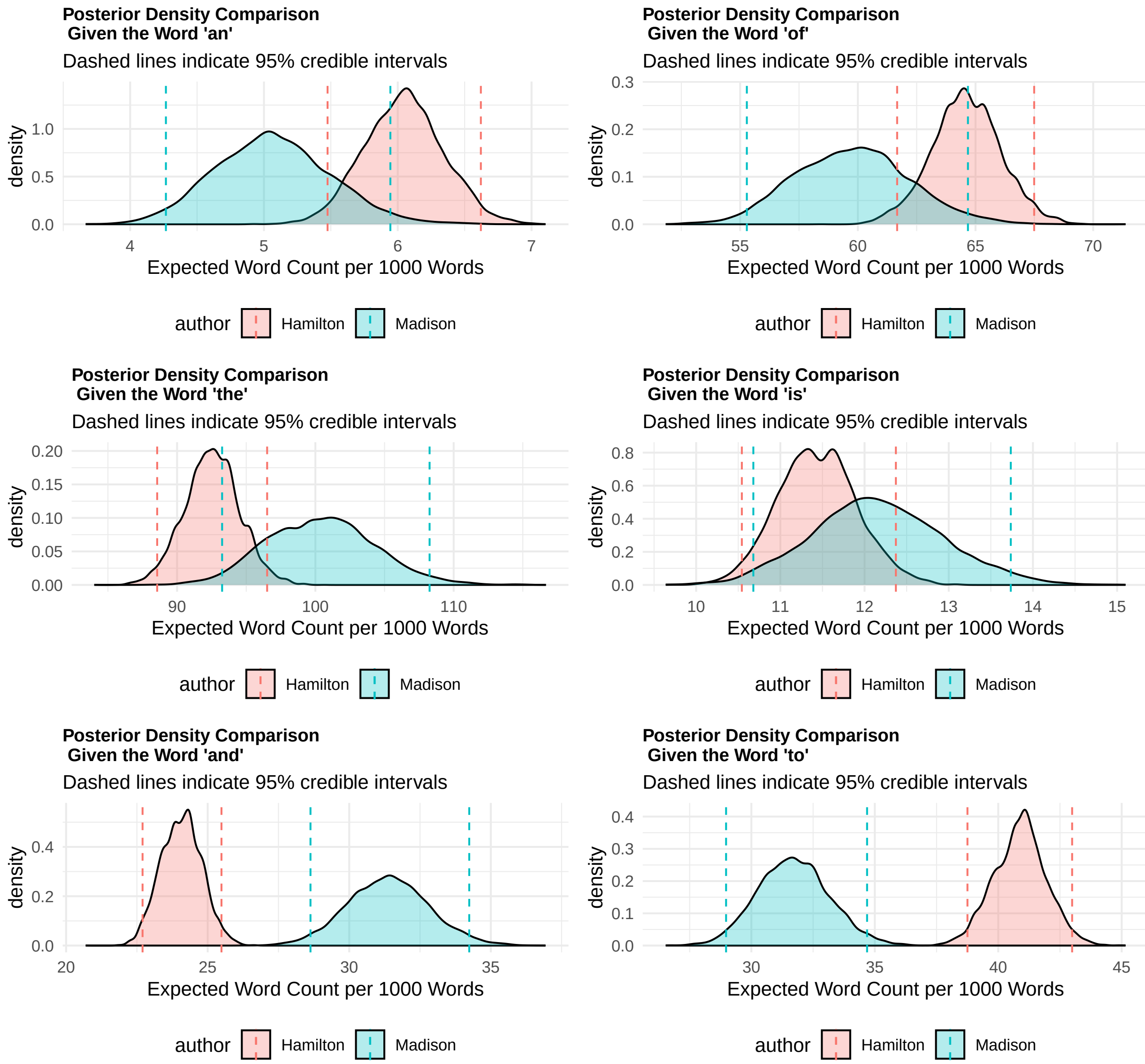

Figure 4. Posterior distributions of filler word usage rates for Madison and Hamilton. The density plots compare author's word frequencies. Words with no overlap in their 95% credible interval are candidates for being influencial in attributing authorship, but words with overlapping 95% credible intervals can provide less determining power.

Table 3: 95 Percent Credible Intervals for Each Author–Word Pair

Word	Author	Lower 95 Percent CI	Upper 95 Percent CI
an	Madison	4.267	5.945
an	Hamilton	5.475	6.621
of	Madison	55.284	64.679
of	Hamilton	61.670	67.492
the	Madison	93.257	108.259
the	Hamilton	88.560	96.512
is	Madison	10.679	13.737
is	Hamilton	10.543	12.372
and	Madison	28.631	34.238
and	Hamilton	22.700	25.487
to	Madison	28.975	34.687
to	Hamilton	38.752	42.992

Shown above is the comparison of posterior densities (Figure 4) between Madison and Hamilton for each word. On each of the six graphs, Hamilton's use of the word is shown in red and Madison's is shown in blue. We are looking to assess a word's usefulness for identifying the author of a Federalist Paper with a disputed author. If we have a word in which the posterior distribution of the use of that word is identical or similar between Madison and Hamilton, then the word is of no use to us because upon an observation of the rate of use of that word in a disputed article, we will struggle to predict which author's distribution of use of that word it came from. On the other hand, if we have a word in which the posterior distributions of the use of that word are very different between Madison and Hamilton, then that word is likely helpful in identifying the author of a disputed article. It would be helpful because we can observe the rate of use of the given word and analyze which author's distribution of use of the word the observation was more likely to come from. Therefore, words in which both authors' distribution of rate of use of the word have very little overlap are likely helpful words for solving the authorship problem. We observe that the words "and" and "to", have very little overlap between Madison and Hamilton's distributions and their 95% credible intervals do not overlap (shown in Figure 4, shown numerically in Table 3). This means that these words could be helpful. On the other hand, "an", "of", "the", and "is" all have moderate to high overlap between the distributions and the 95% credible intervals (Figure 4 and Table 3). This means that we could observe a rate of use of one of these 4 words in a disputed article and have it fall somewhere in the overlap of the credible interval such that the rate of use of the word could have come from either author and we would not be able to discern which author wrote the article. Therefore, "an", "of", "the" and "is" are less helpful words than "and" and "to".

4 Conclusion

We conclude that the words "and" and "to" could be helpful for deciding whether Madison or Hamilton wrote a disputed article, while the words "an", "of", "the", and "is" are probably less helpful. More importantly, we have developed a method to assess the "helpfulness" of any filler word in solving the authorship problem. We do so by generating posterior distributions for the rate of use of each word for Madison and Hamilton (Figure 4) and comparing the distributions and checking for overlap in credible intervals of each distribution. It is perhaps an unreasonable assumption to put the same prior distribution on every filler word. We could make a more informative model if we were to research the rate of use of specific filler words in the English language to inform our prior choices more accurately. In our posterior predictive check, we showed that it is a reasonable assumption to assume a negative binomial sampling model. In order to predict the authorship of the disputed articles, we would first use the method described in this case study to identify the most helpful words. Then, we would observe the rate of use of these words in the disputed articles and analyze which author the use of the words most closely aligns with.